

HOMEWORK 5

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Problem 1. Recall the following theorem from textbook:

If $\{K_\alpha\}$ is the collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty. Give counterexamples showing that the above result becomes false if the word "compact" is replaced by "closed" or by "bounded."

Proof. Replace the definition with "closed." Let $\mathcal{K} = [n, \infty]$, where $n \in \mathbb{N}$. Thus, we construct a finite collection: $\mathcal{K}_1 \cap \mathcal{K}_2 \cap \dots \cap \mathcal{K}_n$. This would be equal to $[\max[\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_n], \infty]$. Hence, the intersection is non-empty. But then we see

$$\bigcap_{n=1}^{\infty} \mathcal{K}_n = \emptyset$$

So closed breaks this theorem.

Now replace "compact" with bounded. Choose $\ell = (0, \frac{1}{n})$, where $n \in \mathbb{N}$. Thus, we see $\ell_1 \cap \dots \cap \ell_t = \frac{1}{n_1} \cap \dots \cap \frac{1}{n_t}$. The intersection is equal to $(0, \frac{1}{\max[n_1, n_2, \dots, n_t]})$. So, finite intersections are non-empty. But we see

$$\bigcap_{t=1}^{\infty} \ell_t = \emptyset$$

So, bounded breaks this theorem as well.

If "compact" is replaced by "bounded and closed," then this cannot be definitely true, because X can be an arbitrary metric space such that $X \not\subseteq \mathbb{R}^n$. In fact, X can be infinite dimensional. This would cause an issue, because Heine-Borel Theorem cannot be applied here, and we see that we need something more to conclude compactness. $\square$

Problem 2. For $x \in \mathbb{R}^1$ and $y \in \mathbb{R}^1$, define

$$d_1(x, y) = (x - y)^2$$

$$d_2(x, y) = \sqrt{|x - y|}$$

$$d_3(x, y) = |x^2 - y^2|$$

$$d_4(x, y) = |x - 2y|$$

$$d_5(x, y) = \frac{|x - y|}{1 + |x - y|}$$

Determine, for each of these, whether it is a metric or not.

Proof. We see $\mathbf{d}_1$ is nonnegative and symmetric. We must show $d_1(x, z) \leq d(x, y) + d(y, z)$. Expanding this, we see $x^2 + z^2 - 2xz \leq x^2 + y^2 - 2xy + y^2 + z^2 - 2yz$. Eliminating terms, we see $-2xz \leq 2y^2 - 2xy - 2yz$, in which we see $-xz \leq y(y - x - z)$. We see if $x = 0$, we let $y = z/2$, and we get $0 \leq \frac{z}{2}(-\frac{z}{2})$, which isn't possible. Thus, $\mathbf{d}_1$ is not a metric.

For $\mathbf{d}_2$, we see that it is nonnegative, symmetric, and when using the triangle inequality, $\sqrt{|x - z|} \leq \sqrt{|x - y| + |y - z|}$ holds. Thus, d_2 is a metric.

d_3 fails. Simply let $x = 1$ and $y = -1$, and we get 0.

d_4 fails. $x = y$ results in a nonzero integer iff $x = y \neq 0$.

For $d_5(x, y) = \frac{|x-y|}{1+|x-y|}$, we have $d_5(x, y) \geq 0$, $d_5(x, y) = 0 \iff x = y$, and $d_5(x, y) = d_5(y, x)$.

For the triangle inequality, let $a = |x - y|$ and $b = |y - z|$. Since $|x - z| \leq a + b$,

$$d_5(x, z) \leq \frac{a+b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b} = d_5(x, y) + d_5(y, z).$$

Thus d_5 is a metric. $\square$

Problem 3. Let K_1 and K_2 be compact subsets of a metric space X .

- Prove that $K_1 \cup K_2$ is compact.
- Prove that $K_1 \cap K_2$ is compact.
- Give an example of infinitely many compact subsets of $\mathbb{R}$ whose union is not compact.

Proof. Let $A = \bigcup_{\alpha}^n U_{\alpha}$ be a finite subcovering of K_1 and let $B = \bigcup_{\alpha}^n I_{\alpha}$ be a finite subcovering of K_2 . Since $K_1 \subseteq A$ and $K_2 \subseteq B$, we may deduce $K_1 \cup K_2 \subseteq A \cup B$. Since both A and B are a finite union of open covers, their union is also finite and open.

The same finite covering shown above can be used for $K_1 \cap K_2$. If you want to shorten the list of how many subcoverings necessary, we simply check the intersection between each subcovering with $K_1 \cap K_2$: Let $p \in K_1 \cap K_2$. If $U_1 \cap (K_1 \cap K_2)$ contains p , then include that subcovering; otherwise, we exclude it from the list of necessary subcoverings. The same applied for I_{α} would yield us a smaller list of subcoverings necessary to cover $K_1 \cap K_2$.

Consider the compact sets $\{[-n, n]\}$. Taking finitely many of them, they are closed and bounded metric spaces, so they are compact. We see

$$\bigcup_{n=1}^{\infty} [-n, n] = \mathbb{R}$$

but $\mathbb{R}$ is not compact. $\square$

Problem 4. Let $K \subset \mathbb{R}^1$ consist of 0 and the numbers $\frac{1}{n}$ for $n \in \mathbb{N}$. Prove that K is compact directly from the definition without using Heine Borel Theorem.

Proof. Let $K = \{\frac{1}{n}\} \cup \{0\}$, where $n \in \mathbb{N}$. Let $\{G_{\alpha}\}$ be a collection of open covers. We see that K is closed, because $K = K \cup K'$, where $K' = \{0\}$, which is contained in K . We see $0 \in G_{\alpha_0}$. Since G_{α_0} is open, $\exists r > 0$ such that $(-r, r) \subseteq G_{\alpha_0}$. We see $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$; thus, $\exists N \in \mathbb{N}$ such that for $n \geq N$, $\frac{1}{n} \in G_{\alpha_0}$. Now, we may list the remaining points: $1, \frac{1}{2}, \dots, \frac{1}{N-1}$. These are finite, so we can dedicate a subcover for each point. Thus, we have $G_{\alpha_0}, G_{\alpha_1}, \dots, G_{\alpha_{N-1}}$, which is a finite subcover. Thus, K is compact. $\square$

Problem 5. Let X be a metric space. Prove that X is compact if and only if every family $\mathcal{F}$ of closed subsets of X with the finite intersection property satisfies

$$\bigcap_{F \in \mathcal{F}} F \neq \emptyset.$$

(Recall that $\mathcal{F}$ has the finite intersection property if the intersection of every finite subfamily of $\mathcal{F}$ is nonempty).

Proof. Suppose $\bigcap_{F \in \mathcal{F}} F = \emptyset$. Taking the complement, we see $X = X \setminus \bigcap_{F \in \mathcal{F}} F = \bigcup_{F \in \mathcal{F}} (X \setminus F)$. Since every $F \in \mathcal{F}$ is closed, the complement is open. So $X \setminus F$ for each $F \in \mathcal{F}$ is an open cover for X . Thus, we see $X = (X \setminus F_1) \cup \dots \cup (X \setminus F_n)$. Taking complements, we see $\emptyset = F_1 \cap \dots \cap F_n$. This is a contradiction because it violates the finite intersection property.

Let $\mathcal{K} = \{U_\alpha\}$ be a collection of open covers. Suppose that there does not exist a finite open cover for X . Taking the complement of these open sets, we see that $\{X \setminus U_\alpha : U_\alpha \in \mathcal{K}\}$ is closed. Let $\mathcal{F}$ be this set. Taking a finite intersection of this, we see $(X \setminus U_{\alpha_1}) \cap \dots \cap (X \setminus U_{\alpha_n}) = X \setminus (U_{\alpha_1} \cup \dots \cup U_{\alpha_n})$. Taking the complement, by our assumption, $(U_{\alpha_1} \cup \dots \cup U_{\alpha_n}) \neq X$, which implies $X \setminus (U_{\alpha_1} \cup \dots \cup U_{\alpha_n}) \neq \emptyset$. With what is given, $\bigcap_{F \in \mathcal{F}} F = X \setminus \bigcup_{\alpha} U_\alpha$. Since $\mathcal{K}$ is an open cover, $\bigcup_{\alpha} U_\alpha = X$. But this implies $X \setminus X = \emptyset$. A contradiction. Therefore, X is compact. $\square$